Ex 37

Angular magnification of a magnifying glass: $\boxed{M = \frac{s_0}{f}}$

refractive power: $D = \frac{1}{f}$ (unit: $dpt = m^{-1}$)

$$\Rightarrow M = s_0 D = (0.25\,m) \cdot (12\,dpt)$$
$$= 3$$

Ex 38

$$\underset{\text{(horizontal)}}{I_1} \longrightarrow \underset{(\theta = ?)}{\text{polarizer}} \longrightarrow I_2$$

$\boxed{I_2 = I_1 \cos^2\theta}$ Malus law

$$\Rightarrow \cos^2\theta = \frac{I_2}{I_1}$$

$$\Rightarrow \theta = \arccos\sqrt{\underbrace{\frac{I_2}{I_1}}_{=0.15}} = 67.2^\circ$$

Ex 39

polarizer 1: θ_1

polarizer 2: $\theta_2 = 45^\circ$ (with respect to θ_1)

polarizer 3: $\theta_3 = 90^\circ$ (wrt. θ_2)

a) True, because $I_3 = I_2 \cdot \underbrace{\cos^2(90^\circ)}_{=0} = 0$

b) True, because $I_2 = I_1 \cdot \underbrace{\cos^2(45^\circ)}_{1/2} = 0.5\, I_1$

c) False, because $I_2 \neq 0$, $I_3 = 0$.

d) True, because the initial light is unpolarized.

e) True, because the relative angle between 1 and 2 is always 45°.

Ex 40

a) For full polarization, we need

$$\boxed{\tan \alpha_B = \frac{n_{glass}}{n_{air}}} \Rightarrow \alpha_B = \arctan(1.4501)$$

$$= 55.41^\circ$$

Brewster's law (507-3)

b) $\boxed{\alpha = \varphi \cdot c \cdot d}$ (507-7)

$$\Rightarrow c = \frac{\alpha}{\varphi \cdot d} = \frac{20^\circ}{(66.5\ \ell \cdot g^{-1} \cdot m^{-1}) \cdot (0.1\,m)}$$

$$= 3.0\ g/\ell$$

c) The opposite chirality variant causes

a $\alpha' = -20^\circ$ rotation.

$$\alpha_{tot} = \alpha + \alpha' = +20^\circ - 20^\circ = 0$$